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What Makes a Powerful Engineering Resume?

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YOUR HOST

Who Am I — and Why Does This Matter?

BACKGROUND



M.Eng. + B.S. Mechanical Engineering

Iowa State University — *your school*



10 years engineering career · 8 at John Deere

4 promotions · 4 discipline pivots



Mech → Embedded Software → Systems → Eng. Leadership

Navigated every transition from inside the same company



Both sides of the hiring table

Applied to jobs · Hired engineers as manager · Interviewed candidates



Systems Engineering Certificate

California Institute of Technology (Caltech)

CLIENT RESULTS

- Helped clients land roles at Fortune 100 companies
- Coached new graduates into their first full-time engineering roles
- Coached entry-level engineers into their first promotions
- Coached engineers into senior and staff-level roles
- Worked across software, hardware, power, systems, and mechanical engineering
- Clients placed at companies across defense, tech, energy, and manufacturing

Disclaimers:

1. This is what has worked for me and others I've coached — your situation may vary.
2. Resume writing is part science, part art.

TODAY'S SESSION

What We're Covering Today

- | | | |
|---|---|--------|
| 1 | Why great engineers don't hear back — the actual root cause | 5 min |
| 2 | The 4-Part Impact Bullet Builder™ Framework — A · M · H · W | 10 min |
| 3 | Your Group's Report Card — Strengths & Gaps from all 8 audits | 8 min |
| 4 | Your Survey Responses — Pain points you submitted | 2 min |
| 5 | ● Live Demo — 3 Bullet Transformations (interactive) | 10 min |
| 6 | The 6 Highest-Impact Changes for This Cohort | 5 min |
| 7 | Open Q&A | 10 min |

THE CORE ISSUE

**The experience is real.
How it's packaged and presented is a
problem.**

"The gap between ★★ and ★★★ is **not a gap in experience or ability.**
It's a gap in translation — and translation is a **learnable skill.**"

50/₁₀₀

GROUP AVERAGE

39

LOWEST SCORE

66

HIGHEST SCORE

7/₈

LANDED IN ★★

THE ROOT CAUSE

Why Great Engineers Get Ignored

X

Bullets read like job duties, not accomplishments

"Responsible for managing the test pipeline" — a duty. Hiring managers see hundreds. They all sound the same.

X

No numbers backing up any claims

"Improved efficiency" means nothing. By how much? Over how many nodes? Across what time period?

X

No detail on HOW you actually did it

Your tools and methods should appear inside the bullet — not just in a skills section at the bottom of the page.

X

The engineering impact is nowhere to be found

Why did the work matter? What did it enable? If a hiring engineer can't tell — the resume goes in the pile.

WHAT YOU'RE EXPERIENCING

The Symptom You All Described

5 of 8 of you wrote the same thing in your survey:

"Not hearing back at all."

This isn't a competition problem.

It's a **communication problem**.

The work is real. The results just aren't making it onto the page.

THE FRAMEWORK

The Impact Bullet Builder™

A 4-part formula for turning engineering and research work into recruiter-readable results

A

Accomplishment

What did you DO — with a result?

M

Metric

A specific number with a unit

H

How / Method

Tool, algorithm, or framework
— named in the bullet

W

Why / Impact

What does it enable, replace,
or make possible?

Formula used by engineers who landed roles at Google · SpaceX · John Deere · Ford Motor Company · GE

A

Accomplishment

Something you **DID with a result** — not a duty, not a responsibility, not an ongoing standing task.

THE TEST

"Could someone else with the same job title have written this exact bullet?"

If yes → it's describing the role, not you. Rewrite it.

★ DUTY

"Responsible for maintaining integration test infrastructure."

★★★ ACCOMPLISHMENT

"Redesigned the integration test pipeline — reducing suite runtime by 62% and eliminating flaky failures across 14 CI environments."

M

Metric

A **specific number** that gives the claim a floor to stand on.

WHAT COUNTS FOR EE/CE/CS RESEARCH

% · dB · ms · accuracy · F1 score · BLEU score
dataset size · node count · throughput (Mbps)
PCE (%) · XRD crystallinity index · entanglement fidelity
kW · publication venue · benchmark rank

NOT METRICS

"several" · "multiple" · "significant" · "improved" · "various" ·
"many"

💡 The number is already in your lab notebook, code output, or poster file.
It just hasn't made it to your resume yet.



How / Method

Name the specific **tool, algorithm, or framework** inside the bullet text — not in the role header, not in the skills section.

EXAMPLES

- ✓ "using GAMS and CPLEX"
- ✓ "via a PyTorch-PennyLane hybrid workflow"
- ✓ "with TensorRT FP16 quantization"
- ✓ "using the MUSIC direction-finding algorithm"
- ✓ "via SEM and XRD characterization"

Your group scored **6.75/10** here — above average. The instinct to name tools in context is right. Keep doing it.

W

Why / Impact

Connect the result to an **engineering purpose**. What does it enable? What does it replace? What becomes possible because of it?

X NOT WHY

"Improving performance" / "Enhancing system efficiency"

✓ WHY — WITH ENGINEERING PURPOSE

- ✓ "Enabling real-time spectrum allocation without centralized coordination"
- ✓ "Enabling planning studies at a fraction of prior computational cost"
- ✓ "Enabling deployment in clinical settings with limited labeled data"

PUT IT TOGETHER

The Full Formula

[Strong Verb] + [What You Built] using [Tool / Method] —
achieving [Metric], enabling [Why / Impact]

★ BEFORE — VAGUE ACTION, NO NUMBER, NO METHOD, NO IMPACT

"Worked on a machine learning model to improve the accuracy of image classification."

★★★ AFTER — ALL 4 COMPONENTS

"Built a convolutional neural network **in PyTorch** for defect detection in manufactured parts — achieving **93% accuracy** on **8,000 test images**, up from a 71% baseline, enabling **automated quality screening without manual inspection.**"

A — Built · improved over baseline

M — 93% accuracy · 8,000 images · 22-pt gain

H — Convolutional neural network · PyTorch

W — Automated production quality screening

★ = 1 component

★★ = 2–3 components

★★★ = all four

YOUR GROUP'S REPORT

Your Group's Report Card

50_{/100}

GROUP AVERAGE ★★

39

LOWEST SCORE

66

HIGHEST SCORE ★★★

7_{/8}

LANDED IN ★★

Scores ranged from **39 to 66 out of 100**. 7 of 8 candidates landed in the ★★ band — solid technical work, but the translation to the page wasn't there yet. One candidate reached ★★★. The difference wasn't research quality. It was how clearly the work was described.

The group spans AI, wireless communications, power systems, semiconductor devices, computer vision, quantum security, and RF design — different fields, same pattern of gaps.

CRITERION ANALYSIS

How Your Group Scored on the Framework

Group averages on the 4 IBB components — scored across all 8 resumes (max 10.0 each)



What is working

The **How** score (6.75) is the group's strength — tools and methods appear inside bullets, not just in the skills section. That instinct is right.

The biggest gap

Metrics (2.88) is the lowest score. Every research domain in this group produces numbers. They just aren't on the page yet.

THE CORE FINDING

The Three Patterns Holding This Group Back

No numbers — 6 of 8 scored 1–2 on Metrics

Every research domain in this group produces measurable results — accuracy rates, efficiency gains, time reductions, signal levels. Those numbers exist in your lab notebooks and code output. They just are not on the page.

No before-state — results appear without context

"Improved performance by 40%" means nothing without a baseline. The strongest bullet in this group worked because it showed both states — reduced a 90,000–element system to 900. Before and after, in the same sentence.

PhD research is the weakest section on every resume — 8 of 8

The most important credential on a PhD resume is consistently described the least clearly. Phrases like "Exploring evaluation frameworks" or "Working on neural network architectures" describe *activities*, not *outputs*. They tell a hiring engineer nothing about what you built, what you found, or what it enables. One candidate's PhD research on a topic directly relevant to every major AI company appeared as a single line in the Education section — zero bullets, zero outputs, zero methods named.

THE DEFINING ISSUE

Your Most Important Credential Is Your Weakest Section

8 of 8 resumes — PhD research is underdeveloped or invisible, across every domain in this group.

WHAT WAS FOUND

- X Research buried as a single line in Education under "Research Focus" — not written as experience bullets
- X Bullets open with: "Exploring..." / "Working on..." / "Investigating..." — these signal activity, not output
- X Publications listed at the bottom of the resume — never connected to the bullet that produced them
- X One candidate's research directly relevant to AI alignment hiring appeared as one sentence. Zero bullets. Zero outputs. Zero methods.

WHAT IT LOOKS LIKE

Real example — from this group's audits

"Working on Neural Network, architecture search, benchmark design, and reliability analysis."

A hiring engineer sees: a list of topics. No tool. No output. No scale. No number. Nothing that tells them what exists now that did not exist before.

THE FIX

Write Your Research Like Your Most Important Job

EVERY RESEARCH BULLET NEEDS THESE 3 THINGS

1. Named approach

What you built, developed, or are using — not just the topic area

2. Scope or scale

How many models, samples, nodes, test cases — give it ground to stand on

3. Result or direction

What it is targeting, what preliminary results show, or what was published. You do not need final results to write this.

Add "with findings published at [venue]" to any bullet tied to a paper.
Four words. Every time.

BEFORE AND AFTER

Before — activity, no output

"Working on Neural Network, architecture search, benchmark design, and reliability analysis."

After — named approach + scope + direction + publication

"Designing evaluation frameworks for reasoning consistency and error behavior in large language models — building benchmark protocols across 6 model families and 4 reasoning task types, targeting failure mode identification for safety-critical deployment, with findings presented at [venue]."

Same research. Same work. Completely different read.

WHAT YOU TOLD ME

Your Survey Responses

"I am not hearing back at all."

— Hadia · Wireless Eng. Co-op applicant

"I struggle to translate PhD projects into clear, concise resume bullets that match what recruiters are looking for."

— Himani · AI/ML, PhD

"PhD research-level projects are often irrelevant for industry — how do you cater to this problem?"

— Zaid · Robotics, PhD

"Not hearing back, or getting rejected due to high competition."

— Zohreh · Power Systems internship applicant

"Not hearing back from employers after applying — not even getting a reason."

— Sibtain · AI Engineer (LLM), PhD

All of these trace to the same root cause: **the results aren't stated on the page.**
Not a competition problem. Not an experience problem. A translation problem.

ACTION PLAN

The 6 Highest-Impact Changes

1 Add 1 number to every research bullet

The number already exists — in your code output, lab notebook, poster, or experiment log. You do not need to invent it. Just write it down.

2 Treat your PhD research section like your most important job

It is. Write it as a role with bullets — not as a topic list in the Education section. Named approach + scope + result or direction.

3 Connect every publication to the bullet that produced it

Add "with findings published at [venue]" directly in the experience bullet. Four words. Do it for every paper.

4 Give every major contribution a before-state

"Improved X" has no weight without a baseline. The group's strongest bullet worked because it showed before and after in the same sentence.

5 Convert present participles to outputs

Replace "Exploring..." / "Working on..." / "Investigating..." with what the work has produced — even if it's preliminary. "Developing X — with initial results showing Y" is always stronger.

6 Write your industry experience like it matters

Internships, co-ops, and full-time roles are a differentiator for PhD candidates. Dates-only entries are worse than omitting them. Every industry role needs bullets.

LIVE DEMO

10–Minute Transformation Exercise

3 bullets. Patterns I found across your group's resumes. Let's fix them live.

How This Works

- Step 1: I show the BEFORE bullet
- Step 2: We diagnose — what's missing?
- Step 3: Press → to reveal the AFTER, tagged by A · M · H · W

The 3 Domains

- Demo 1 — Wireless / RF Research
- Demo 2 — AI / ML Pipeline
- Demo 3 — Power Systems / Circuit Design

Press → on each demo slide to reveal the After panel

DEMO #1 — MACHINE LEARNING RESEARCH

Image Classification — Applying the Formula

BEFORE — MISSING METRIC, HOW, AND WHY

"Developed a deep learning model to improve accuracy for image classification tasks."

WHAT'S MISSING?

The verb "Developed" is a solid start — but **no number** ("improve" by how much?), **no method named** (what kind of model?), **no impact** (where does this get used?).

AFTER — ALL 4 COMPONENTS

*"Developed a **convolutional neural network in PyTorch** for defect detection in manufactured parts — achieving **93% accuracy on 8,000 test images**, up from a 71% baseline, enabling **automated quality screening without manual inspection on the production line**."*

A — Developed · improved over baseline

M — 93% accuracy · 8,000 images · 22-pt gain

H — Convolutional neural network · PyTorch

W — Automated production line quality screening

DEMO #2 — CIRCUIT DESIGN / POWER SYSTEMS

Circuit Optimization — Applying the Formula

BEFORE — VAGUE ACTION, NO NUMBER, NO TOOL, NO IMPACT

"Designed and tested a circuit to reduce power consumption in embedded systems."

WHAT'S MISSING?

No number — power reduced by how much? **No tool named** — what software or bench equipment? **No impact** — what does lower power consumption actually enable?

AFTER — ALL 4 COMPONENTS

"Designed a low-power voltage regulator using LTspice simulation and bench testing — reducing idle power draw by 38% vs. the baseline design and extending estimated battery life by 14 hours, enabling deployment in field sensors with a 3-year battery target."

A — Designed · reducing power draw

M — 38% reduction vs. baseline · 14-hr battery gain

H — LTspice simulation + bench testing

W — Field sensors with 3-year battery target

DEMO #3 — SOFTWARE / DATA ENGINEERING

Data Pipeline — Applying the Formula

BEFORE — ACTIVITY DESCRIPTION, NO RESULT

"Worked on a data processing pipeline to improve the speed of the data analysis workflow."

WHAT'S MISSING?

"Worked on" signals activity, not output. No number on how much faster. No tool named. No reason why the speed mattered. Could have been written by anyone in any data role.

AFTER — ALL 4 COMPONENTS

"Rebuilt a data processing pipeline in Python using parallel processing and vectorized operations — reducing analysis runtime from 4.5 hours to 22 minutes on a dataset of 2.3 million records, enabling same-day results for field engineers who previously waited overnight."

A — Rebuilt · reducing runtime

M — 4.5 hr to 22 min · 2.3M records

H — Python · parallel processing · vectorization

W — Same-day results for field engineers

PHD TO INDUSTRY TRANSLATION

Writing PhD Experience for Industry

This is a **translation problem, not a content problem**. Your research produced real engineering outputs — they just need to be described in language that a hiring engineer can parse in 6 seconds.

1. REPLACE THE RESEARCH GOAL WITH THE ENGINEERING PROBLEM

Academic framing

"Investigated the thermal properties of novel polymer composites under cyclic loading conditions."

Industry framing

"Existing thermal materials failed above 150°C under mechanical stress — a gap blocking adoption in electric vehicle battery designs."

2. REPLACE "STUDIED / EXPLORED" WITH WHAT YOU BUILT OR PRODUCED

Academic framing

"Investigated machine learning approaches for anomaly detection in industrial sensor data."

Industry framing

"Built an anomaly detection model on 14 months of industrial sensor data — achieving 91% precision at a 0.3% false positive rate on the held-out test set."

Two More Rewrites — and the Checklist

3. ATTACH YOUR RESULTS DIRECTLY TO THE TECHNIQUE USED

Academic framing

"Characterized thin film morphology using scanning electron microscopy and atomic force microscopy."

Industry framing

"Used electron and atomic-force microscopy to resolve domain sizes of 18–24 nm with surface roughness below 0.4 nm — establishing parameters that guided deposition optimization."

4. END WITH WHAT IT ENABLES — NOT WHAT IT CONTRIBUTES

Academic framing

"...contributing to the understanding of charge transport in perovskite solar cells."

Industry framing

"...identifying the doping level that maximized carrier mobility without accelerating degradation — informing process conditions for stable, high-efficiency device fabrication."

5-QUESTION CHECKLIST BEFORE SUBMITTING ANY PHD BULLET

1. **Context:** Does it say what was broken or unavailable before?
2. **Accomplishment:** Does it name what you built, fabricated, modeled, or ran?
3. **Metric:** Does it include at least one number from your actual results?
4. **How:** Does it name the specific tool or technique used?
5. **Why:** Does it say what an engineer or company can now do because of it?

If any of the five are missing, the bullet is not finished yet.

WHAT COMES NEXT

Take This Further

YOUR INDIVIDUAL AUDIT

Your personal audit report was delivered to the email you submitted. It includes your score, criterion-by-criterion breakdown, and the top-priority bullets to rewrite first.

REFERRAL BONUS

\$200 cash for any club member who refers someone to the Engineering Career Accelerator. Make sure they mention your name at booking.

THE ENGINEERING CAREER ACCELERATOR

3-Session 1-on-1 Coaching Program

- Session 1 — Full resume rebuild using IBB™
- Session 2 — Interview story mastery + story vault
- Session 3 — Live mock interview + real-time feedback

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THE CORE MESSAGE

**"The research is real.
The publications are real.
The technical depth is real.**

**The gap is the translation —
and that's a learnable skill."**

Mansour Manci
Engineering Career Coach

LaunchToLead.io
Free strategy call available

LinkedIn
/in/mansourmanci

Thank You — Open Q&A